

## HOW WORKS A VOLKSWAGEN

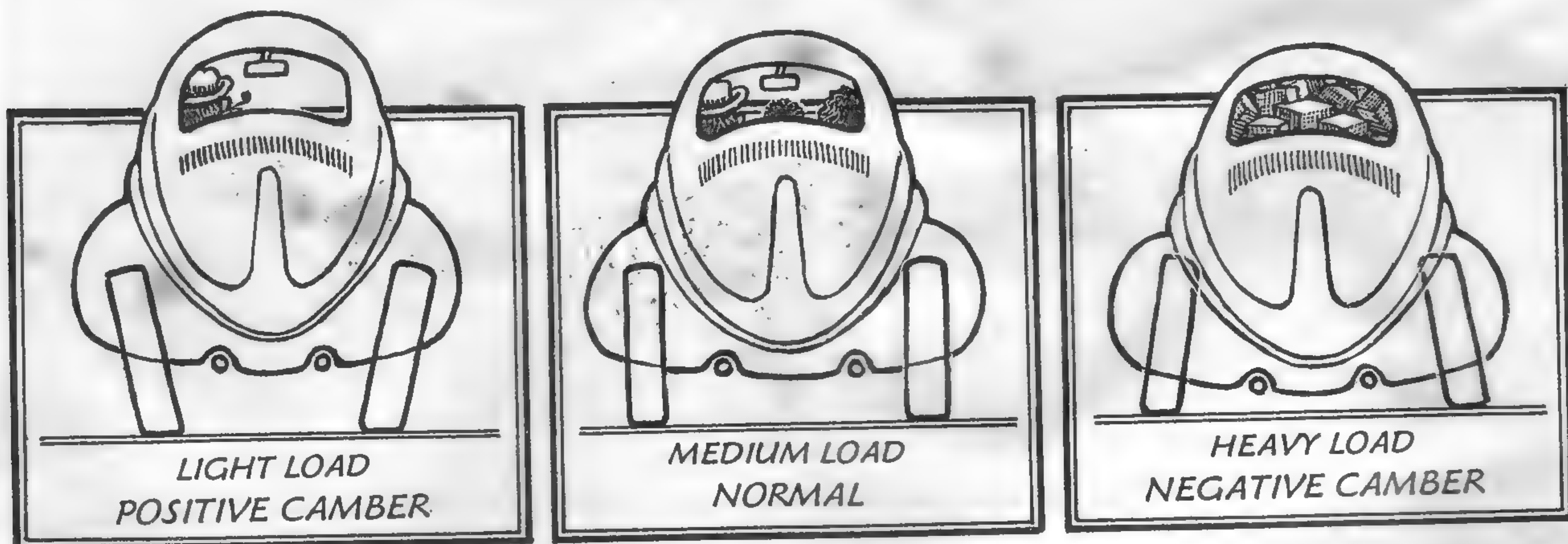


Along with word problems in Math and Zen Principals, the operation of a motor car seems to be a first water mystery. One definition of a mechanic is someone who looks at an automobile as a system. In this frame of reference an idiot would be one who looks at a car as a series of problems. Here is a generalized story to help you put a little light on how a car runs so you can dig it.

Note that front always means the *front of the car*.

The wheels are important, all four of them, plus a spare in case one of the tires goes flat. In the Volkswagen, all four are free to bounce up and down with the vagaries of the road—individually suspended. The front two are mounted on hinged axles, or king pins, so they can turn to give the car direction. The geometry is complicated but the principle is easy. There is caster, camber, and toe-in but they combine to give the best angle between the wheels and the road for maximum control, ease of handling and comfort with minimum of tire wear. The rear wheels apply the force to the road that makes the car go. All four wheels are attached to the **frame** (body) through **torsion bars**—those tubes which run across the front and back of the car—which return the car to its proper level after each road irregularity. The **shock absorbers** (vertical round tubes) merely dampen the up and down movement for rider comfort and safety.

The rear wheels of the Volkswagen have an attitude which is peculiar to cars with individually suspended driven wheels. The torsion bars are set for a normal position with a normal load so the attitude of the wheels is dependent on the amount of weight in the car:



The steering force is applied by you to the **steering wheel** inside the car and is transmitted through a hollow rod to the **steering box** mounted on the frame which changes the direction of the force to an arm which pushes and pulls the **tie rods** and **tie rod ends** and so changes the direction of the wheels.

The **horn** is an electro-magnetic device designed to make noise and act as a warning signal. The hot wire from the battery goes directly to the horn. The ground wire, to complete the circuit, runs up through the hollow steering rod to the horn button.

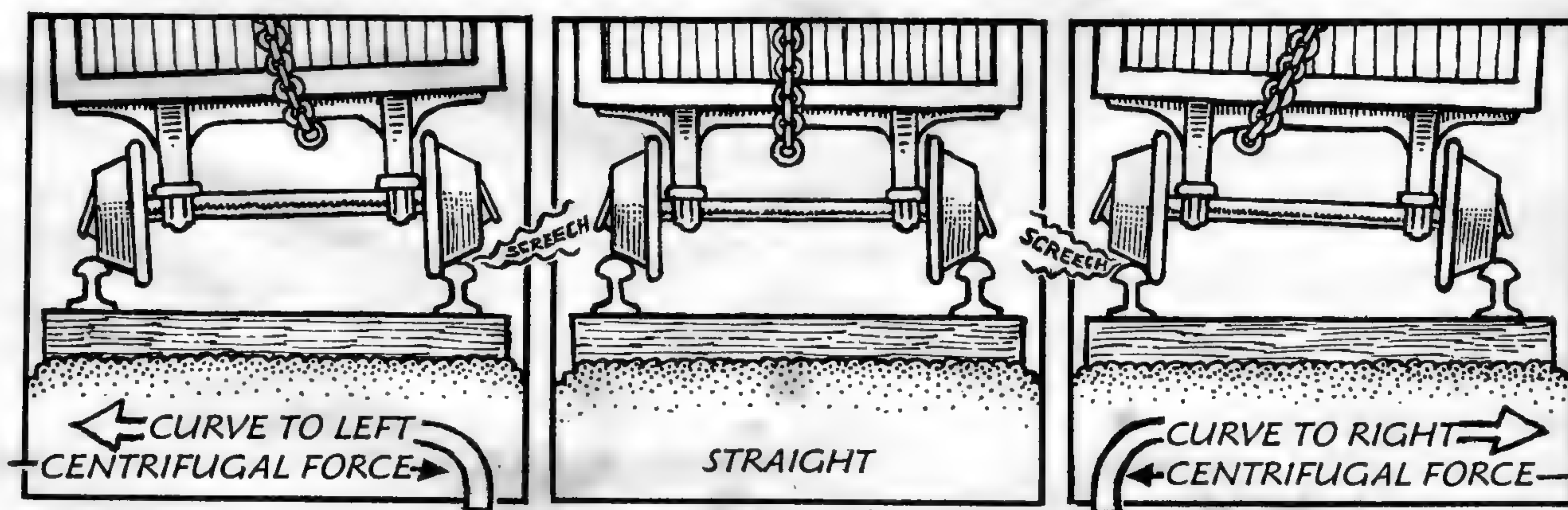
The **brakes** apply negative acceleration to stop the car. The **brake drums** are part of the wheels and the stopping force is friction applied to these drums by **brake linings** on **brake shoes** which are hinged at one point and pushed apart at the opposite point by a hydraulic **wheel cylinder**. These wheel cylinders are slave cylinders and are operated by the **master cylinder** which is moved by the **brake pedal** inside the car. When you push your foot down on the brake pedal, you move the master cylinder which pushes on the brake fluid and that movement is transmitted, evenly, to the four wheel cylinders and the brake lining is shoved against the drums to stop the wheel and thus the car. OK?



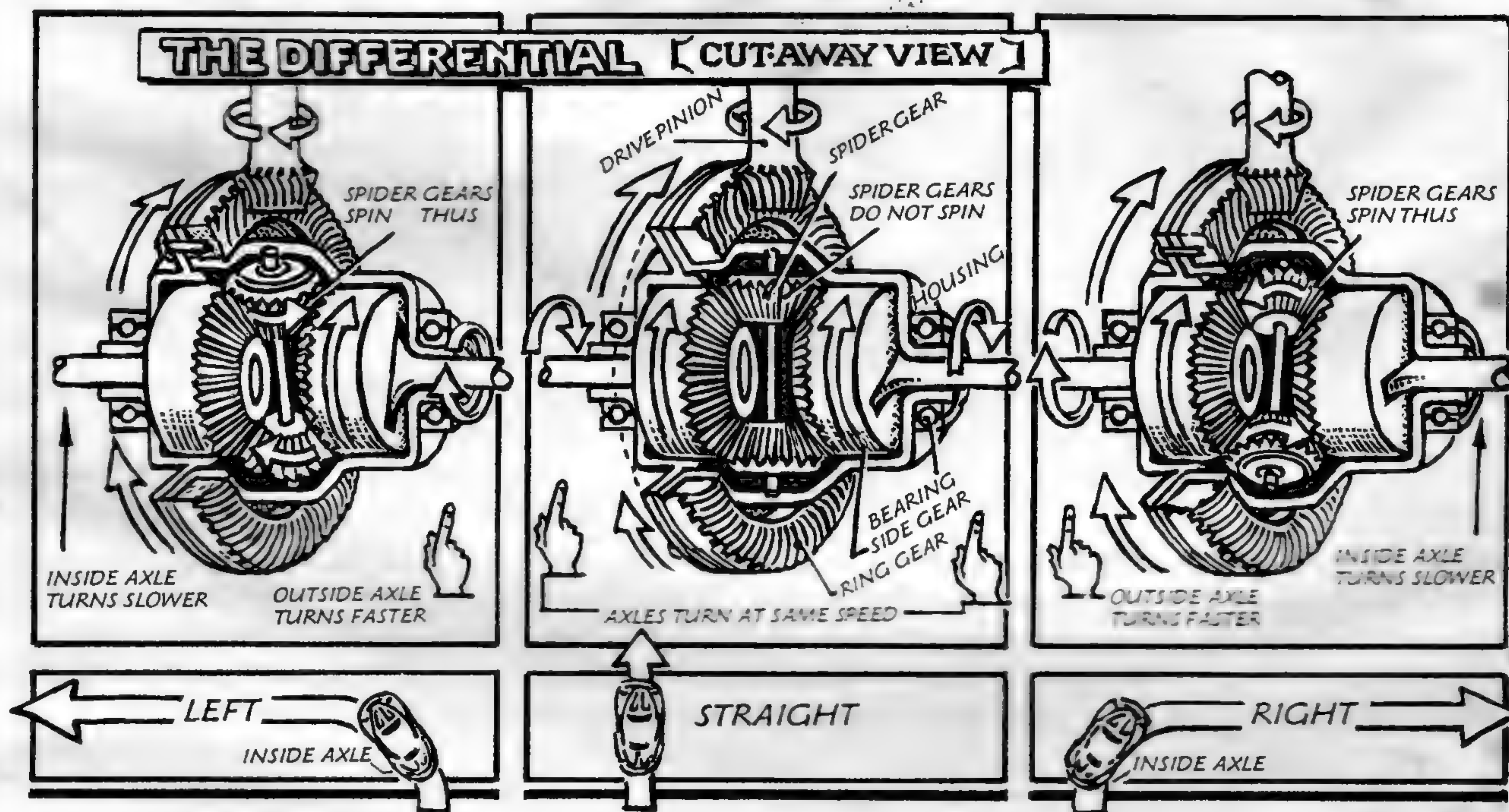
Disc brakes are found on the front wheels of late model Types II, III and IV. These have a disc which is part of the wheel and a caliper which is bolted to the car. A double (dual) master cylinder is used to transfer your foot force to the hydraulic fluid. One section operates the rear drum brakes and the other pushes the brake pads out of the caliper to apply friction to the discs.

The emergency brake is a pair of cables which pull the front shoes of the rear brakes into contact with the drums. The handle in the car has notches and a release to hold and let go the tension in the cable and therefore the braking action. Don't drive with it on.

The two rear wheels are powered by swing drive-shafts which extend to each wheel from the differential, a device used to allow one wheel to travel at a different speed from the other. This is needed to go around curves as the inside wheel on a curve travels a shorter path than the outside. It is well known in human endeavor that we try to make the outside like the inside for sanity but here the inside is obviously not possibly the same as the outside, ergo the internal adjustment—in a car, the differential. The railroad people ingeniously use a flanged, tapered wheel and centrifugal force to enable them to use a solid axle between the wheels, so:



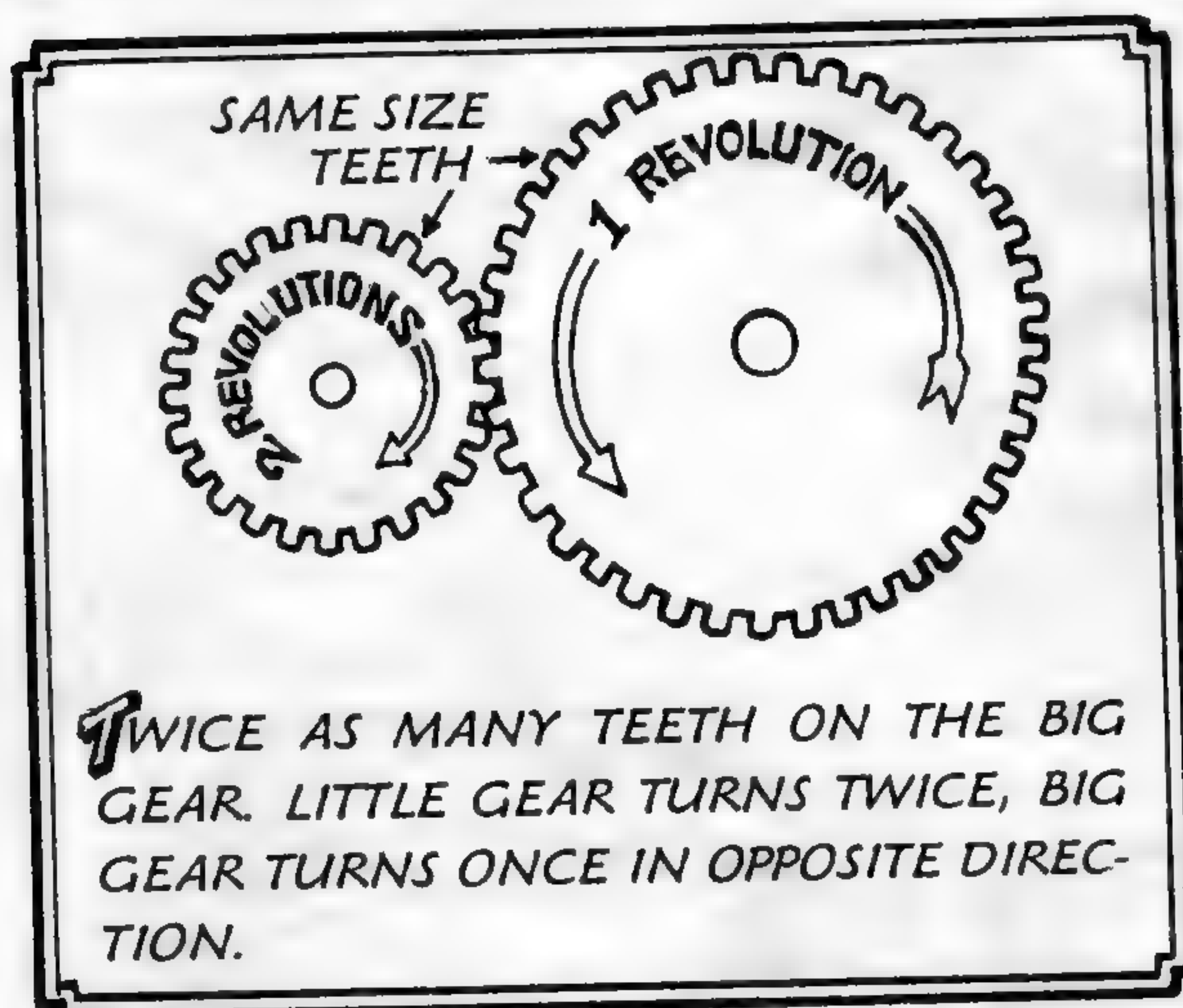
The automobile engineers use the differential, so:





The differential also changes the direction of the twist from the round and round of the motor to the round and round of the wheels. For this it uses the **ring and pinion gears** as shown. Actually, in the Volkswagen, the axles also move up and down with the road and they are spade-ended shafts that fit into sockets which let them operate in separate planes. Late models have shafts that bolt to the differential and use two **universal joints** to transfer the power to the wheels.

The **transmission** is a box containing many gears, shafts, retainers, shifting mechanisms and is complicated in the extreme, but operates on the simple principle that a larger gear will make less revolutions when driven by a smaller gear (with the same size teeth), so:



In the VW, high gear is arranged so the twist from the engine makes almost the same number of revolutions going out as when it came in. The pinion gear shaft comes out of the transmission and drives the ring gear and these two gears have a ratio of about four to one, so the wheels are actually turning about one fourth the revolutions of the engine when the transmission is in high. This isn't exactly correct but I'm telling you a story, not making an engineering analysis. If engines put out the same twist at all revolutions, the problem wouldn't exist but the Volkswagen engine works best between 2,000 and 3,500 revs per minute. The gears in the transmission increase the ratio between the engine and the turning of the rear wheels in the lower or first gear and about the same in reverse. In second the engine turns about eight times as fast as

the wheels and in third about six times and, as we have said, about four times the wheel revolutions in high. With the engine turning a constant 3500 rpm, the car will be going about 10 miles per hour in first, about twenty in second, about thirty-five in third and about fifty-five in high or fourth. The transmission is then a gearing device to match the speed of the engine (relatively constant) with the speed and power requirements of the rear wheels. The old Bus has an extra set of straight gears out by the wheels which adds ground clearance and increases the ratio of the engine speed by one and a quarter to one, which slows the Bus a little, but also adds power at the wheels for extra load-carrying capacity.

Late model Busses no longer have these extra gear boxes at the rear wheels but use **double jointed swing axles**, thus gaining higher highway speed and losing a lot of noise and some power. Reverse twist to move the car backwards is obtained by using an **idler gear** to change the direction of the twist.

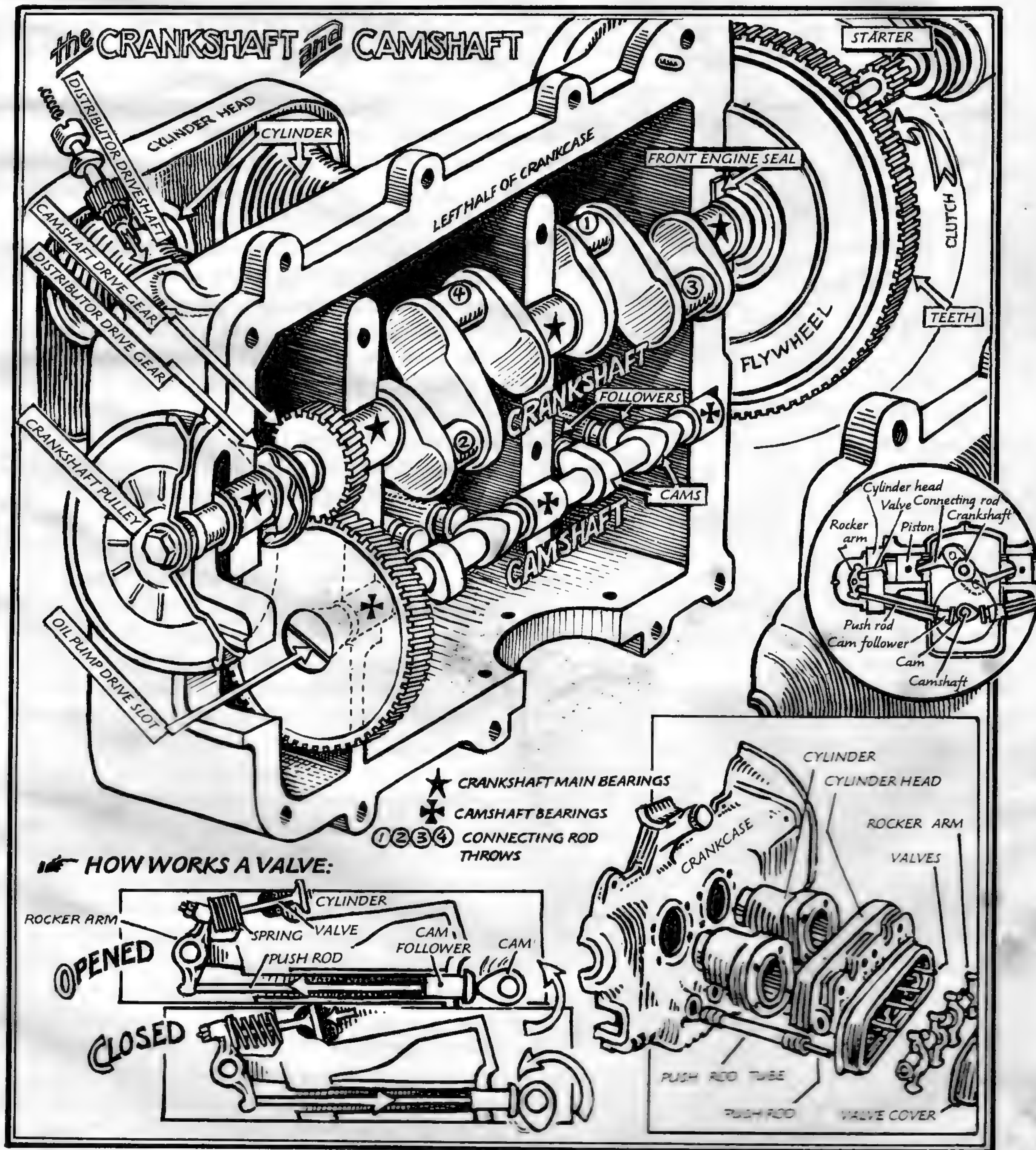
The transmission is connected to the engine through the **clutch**, a spring-loaded friction device which connects and disconnects the engine from the car for starting, idling and shifting. The **flywheel** provides one face and the **clutch assembly** provides the other with the two-surfaced **clutch plate** between. The clutch assembly is spring loaded and presses the clutch plate between its surface and the surface of the flywheel to make the connection and you know how important that can be. When you push on the **clutch pedal** in the car, you are actually compressing the springs in the clutch assembly through a lever and the clutch release bearing and letting the clutch plate run free, effectively disconnecting the engine from the transmission.

The **flywheel**, a heavy machined chunk of steel, serves a triple purpose. The machined face acts as part of the clutch. There are teeth all around the outer circumference which the starter assembly engages to turn the engine over for starting. It also serves as a heavy rotating body bolted to the end of the crankshaft to maintain the revolving inertia of the engine through its cycles.

The crankshaft (a heavy steel forging) is fastened to the flywheel with four pins and a gland nut. It runs in four bearings, called **main bearings**, which are massive and pressure-lubricated. The



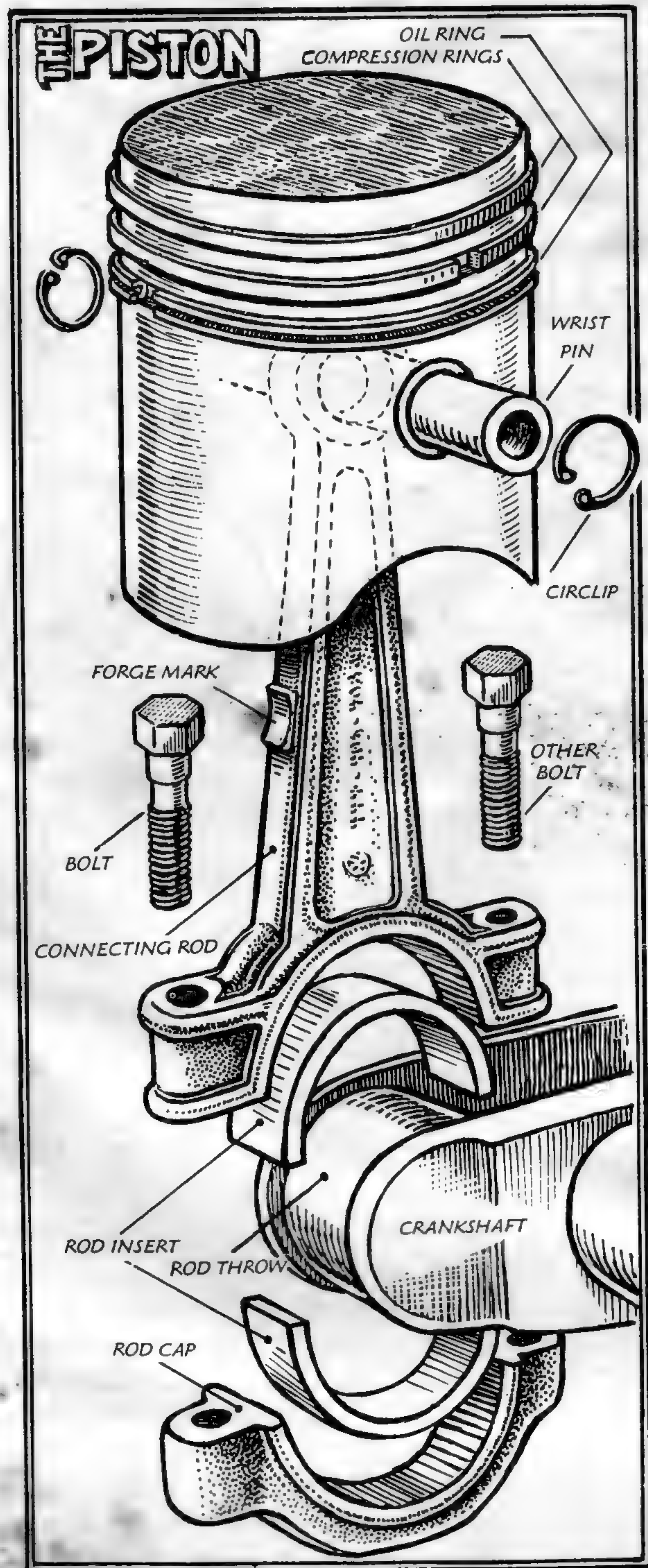
crankshaft has four cranks, or throws, which serve to translate the back and forth movement of the pistons, through the connecting rods, into the round and round motion of the crankshaft and fly-wheel assembly. The connecting rod bearings are also pressure-lubricated. Both the main and connecting rod bearings are thin shells which can be replaced as they wear. The crankshaft assembly looks so:



The crankcase is a split aluminum casting machined in the proper places, which contains that portion of the engine known as the **bottom end**. The main bearings and the cam shaft bearings run through the case so that one half of the bearing is on each part of the case. The cam shaft is geared to the crankshaft and has cams or lobes which operate the cam followers which in turn push the push rods which operate the valves:



The crankcase also acts as an oil sump to collect, store and help cool the oil that is pumped through the bearings by the oil pump. The crankshaft has a funny sideways gear, called a worm gear, pressed on the shaft next to the cam shaft drive gear. This worm gear turns the distributor drive gear which has a slot on top that turns the distributor and an off-center bump which operates the fuel pump through a push rod. The **oil pump** fits into a slot in the end of the cam shaft and turns with it. In the older models, the cam shaft bearings are merely machined into the crankcase but in the later models, they have installed bearing shells to take the wear. On the opposite end of the crankshaft from the flywheel, outside of the crankcase, is attached the **crankshaft pulley** which drives the **generator or alternator** with a belt.



The **pistons** move back and forth in the cylinders and are sealed to the machined sides of the cylinders by the **piston rings**. The pistons are attached to the connecting rods by the **piston pins** which go through the pistons, so:

The **cylinder heads** fit over the cylinders and contain the **valves** and **valve train** which consists of the push rods from the bottom end and the **rocker arms** which push the valve open. The valves are returned to the closed position by the valve springs. There are two valves to each cylinder, one intake and one exhaust, and they are operated by the valve train to let the fuel mixture into the cylinder (intake) and then let the burned gases out after the cylinder is fired (exhaust). The heads and cylinders are held tight to the crankcase by long studs which are screwed into the crankcase. The head nuts tighten this assembly down to the crankcase.

The Volkswagen engine is a four cycle engine which means each cylinder fires once every two revolutions, that is, each cylinder requires four cycles or half revolutions for power production. Let's start with the gas in the tank and go through the complete schmutz. The liquid gasoline is sucked into the fuel pump and then pushed into the carburetor by a neoprene **diaphragm** which is pushed up and down by the rod from the cam shaft. The sucking-pushing action is obtained by wafer valves in the pump: one opens to suck and closes to push and the other opens to push and closes to suck. The gasoline is thereby pushed into the carburetor bowl. There is a float in the carburetor bowl which closes a float valve when the carburetor is full. From the bowl, the gasoline flows into the accelerator pump which is used to pump liquid gas into the carburetor throat in a spray and is operated by the **accelerator pedal**. The gasoline also flows from the bowl to the jets. There is a jet for idling, one for middle speed and one for high speed. These jets are just calibrated holes where liquid gasoline exists, ready to be sucked into the engine as demanded. There is a **butterfly valve** at each end of the carburetor. The top butterfly is the choke and when closed it limits the air intake to the carburetor throat, thereby making the gas-to-air mixture richer in gasoline. The butterfly valve at



the bottom of the carburetor opens the throat to the engine and is attached to the accelerator pedal which controls the speed of the engine. The **accelerator pump** is used to add the necessary extra liquid gasoline when the pedal is pushed hard for extra power. Now we have the gasoline available at the throat of the carburetor and a method of controlling its flow.

1975 and later Types I and II and some Types III and IV have no carburetors. The fuel is supplied directly to the area behind the intake valves, atomized and ready to mix with the incoming air by a system called **Electronic Fuel Injection** (Types III and IV) or **AFC Fuel Injection** (Air Flow Control, Types I and II). These systems use an **electronic fuel pump** to suck the gasoline through a **filter** and pump it into a circular system at about 28 pounds per square inch (psi). The unused gasoline is returned to the tank. The pressure is regulated by a **pressure regulator** to keep a constant supply of 28 psi fuel ready to be injected. The **injectors** are atomizing valves operated by solenoids which open and close at the direction of the **Electronic Control Unit** (brain). This control unit gets messages from various sensors, digests them, then opens the injectors for a specific amount of time, thus injecting atomized fuel right in back of the intake valves ready to be sucked into the cylinders. Although this brain has about 300 (electronic) or about 80 (AFC) diodes, transistors, resistors and condensers, it does a simple task: it determines the amount of time each injector is open, thus making the amount of fuel fit the existing conditions exactly. In order to do its job, the control unit needs to know the temperature and pressure conditions it is going to inject fuel into, OK? It is connected to the **crankcase or manifold temperature sensor**, the **head temperature sensor**, the **pressure switch** (electronic only) and **pressure sensor** (electronic only) and finally to the **timer** on the distributor. There are other components for cold starts and rapid acceleration and electric relays to supply a constant voltage to the fuel pump and control unit. However, under steady-state conditions, the factors are: temperature, pressure and timing.

As soon as the engine is warmed up and running at a fairly steady rate, the temperature sensors, which tend to put in extra fuel for a richer mixture, get out of the action. Then what's left is the timing and the pressure. The timing is provided by a double contact switch under the distributor which merely tells the control unit which two injectors (electronic) or which injector (AFC) to open. This trigger contact is a simple cam and contact system which tells the control unit which cylinders are ready for fuel.

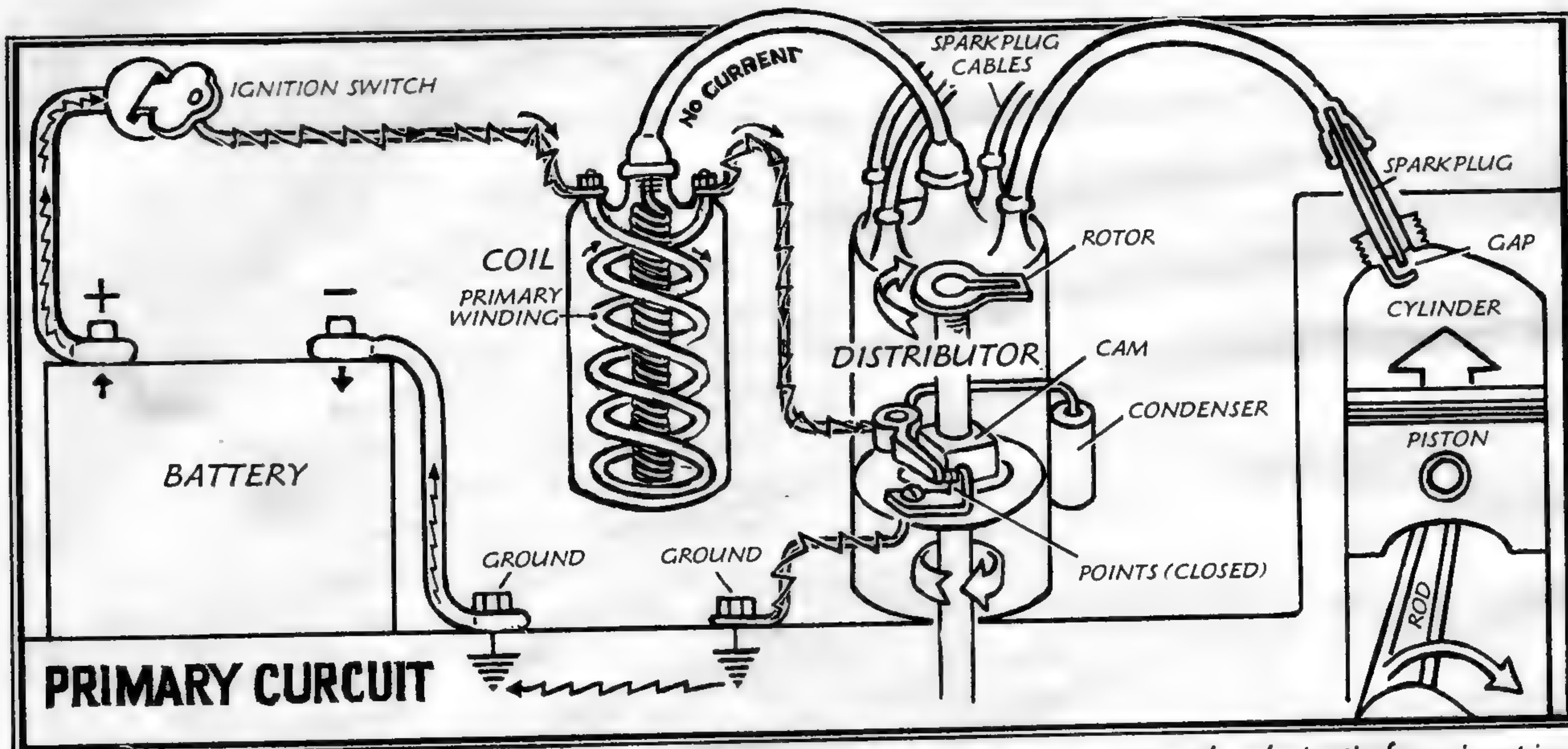
Here, the electronic and AFC systems differ, so we'll start with the **electronic** and explain the **pressure switch** and **pressure sensor**. The accelerator is connected by a cable to the **throttle valve** which is a butterfly valve in the **intake air distributor**. When you push on the accelerator, the butterfly valve opens allowing air from the air cleaner to enter the intake manifold. The pistons are moving up and down in the cylinders sucking air into the intake system and pushing it out through the exhaust system, so the pressure in the manifold is usually less (negative pressure) than outside in the world. The amount the butterfly valve is open (foot pressure on the accelerator) determines the air pressure in the manifold—all right so far?—and the pressure sensor connected to the manifold measures this air pressure and sends the dope along to the control unit which then applies the juice to open the injectors which spray the fuel in back of the intake valves to mix with the air and the mixture is sucked into the cylinder when the valve opens.

When the engine is at idle, the throttle valve (the butterfly valve in the intake air distributor) is closed and air for idle is provided by the **auxiliary air regulator**. The **pressure switch** was used on early models to provide extra fuel at full throttle. This job was next given to an extra circuit in the pressure sensor and now is done by a new throttle valve switch (1972-'74).

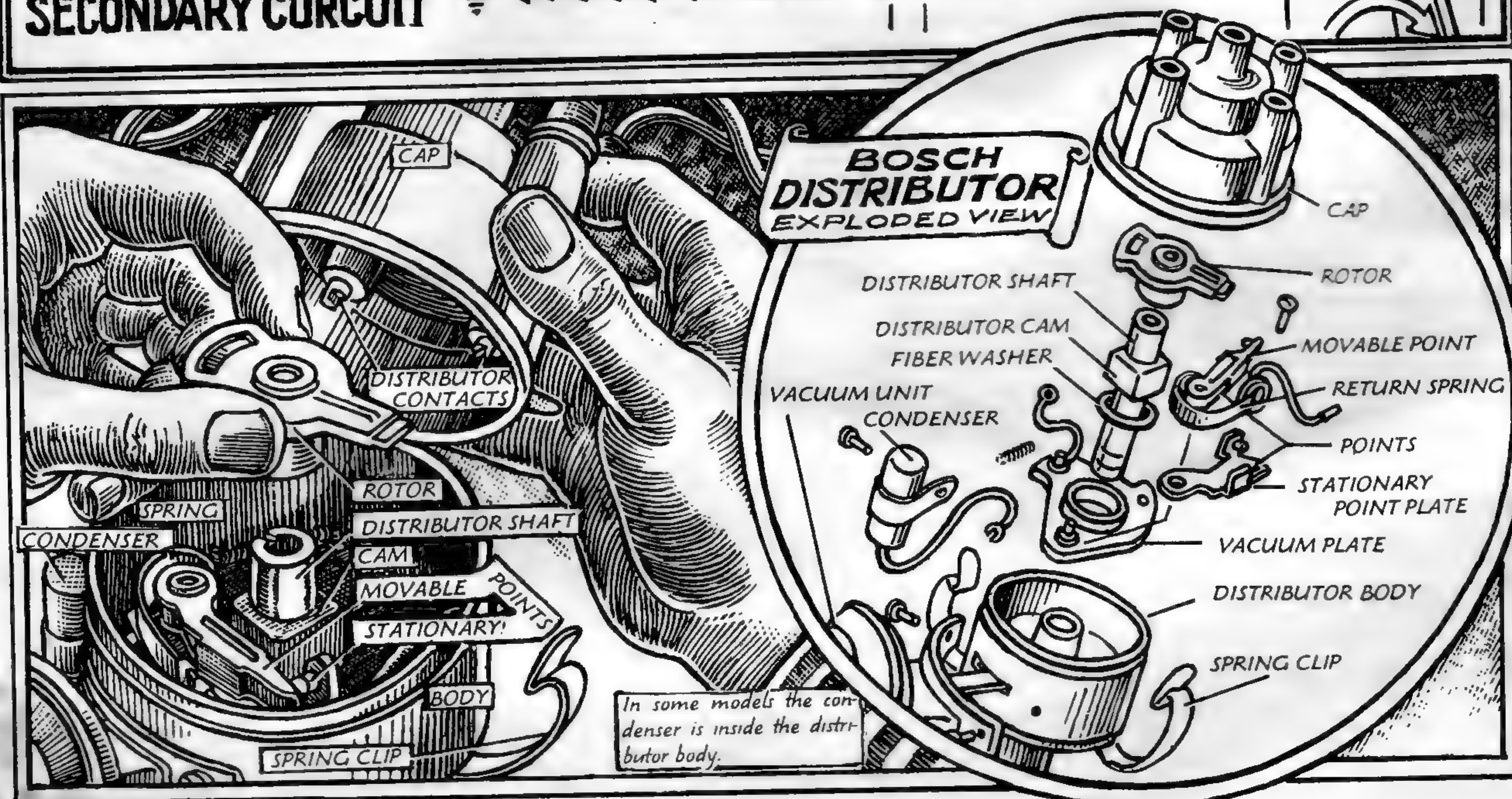
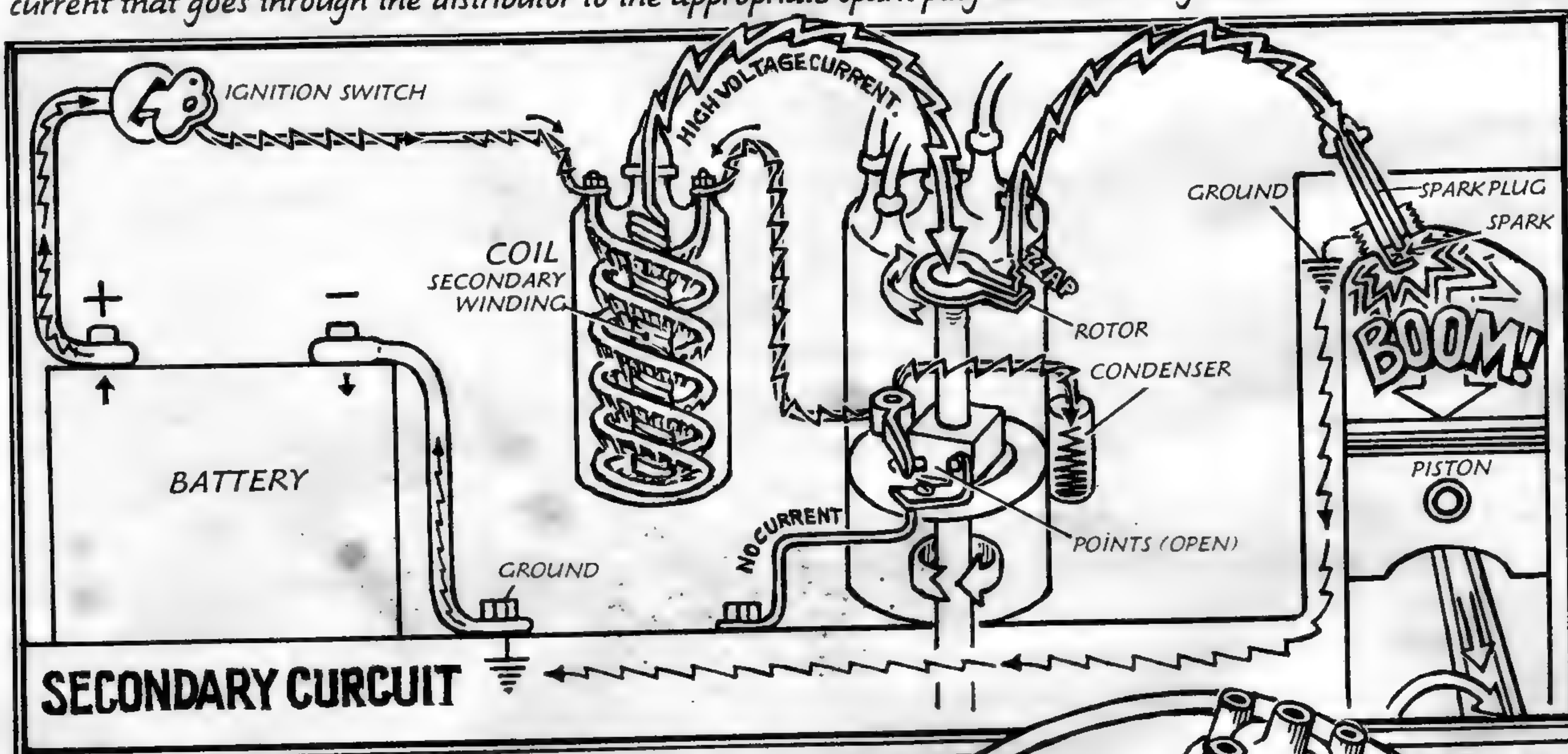
With **AFC** fuel injection when you push on the accelerator, the air flap opens which sends a message directly to the control unit which regulates the amount of fuel sent to the cylinders by the injectors. This system is a lot simpler than the older electronic one.







When the points are open, the condenser stops the primary circuit, preventing the electricity from jumping the points gap and going to ground. The coil responds to this circuit break by producing a high voltage (secondary) current that goes through the distributor to the appropriate spark plug and hence to ground.





Whether fuel injected or carbureted, we need a method of setting this gasoline-air mixture on fire in the cylinder and that's where the ignition system comes in. Let's start with the wire that comes from the battery and goes to the key. When the key is turned all the way, part of the current is used to go to the ignition system and part to the solenoid on the starter. The solenoid is an electromagnet which, when energized, pushes the starter gear into engagement with the teeth on the flywheel and also closes a contact which draws current from the battery through a much heavier wire to turn the starter motor and thus the engine. As soon as the motor starts the key is relaxed and the solenoid drops the starter gear back away from the flywheel.

The other portion of the current from the key goes to the **ignition coil** which is a cylinder hanging on the fan housing. It has three connections, two small ones and one big one in the center. The current from the key goes to one of the small bolts and then out through the other bolt connection to the distributor. This circuit is called the primary and puts six (or twelve) volt current in a coil in the ignition coil. Inside this primary coil are thousands of turns of wire making up the secondary coil which increases the voltage to something around ten thousand volts, which is now available at the center large outlet and connects to the center large connection at the distributor cap. It takes a lot of voltage to make a spark jump and this is the way we get it.

You are starting to see that we are going to burn our gasoline-air mixture with a spark and for that we need **spark plugs**, one for each cylinder and a heavy wire from each plug to its corresponding hole around the distributor cap.

The **distributor** is a device to time the spark impulses to the spark plugs so they fire at the proper time, to burn the mixture, to give the power. The **distributor shaft** is driven round and round by a funny gear on the crankshaft. As it goes round and round inside the distributor it opens and closes a set of points. The distributor shaft is almost square and at each point of the square it opens the **ignition points** and in between the high places allows the ignition points to close.

When the points are closed, the 6 (or 12) volt current from the battery to the key (ignition switch) goes through the primary winding in the **coil** through the closed points, to ground. Electricity, like water, will follow the path of least resistance and if one path is closed, it will follow another. A small cylinder, called the **condenser**, located either outside or inside the bottom of the distributor, is part of this path while the points are open (not touching). The condenser stops the 6 (or 12) low voltage current and keeps it from jumping the point gap and going to ground. This stopping of the current also prevents the points from burning and pitting too rapidly.

When the primary circuit is broken by the condenser, the secondary winding in the coil builds up voltage (pressure) and this high voltage electricity flows from the coil to the center of the distributor cap to the rotor and jumps the post in the cap to the proper wire to the proper spark plug.

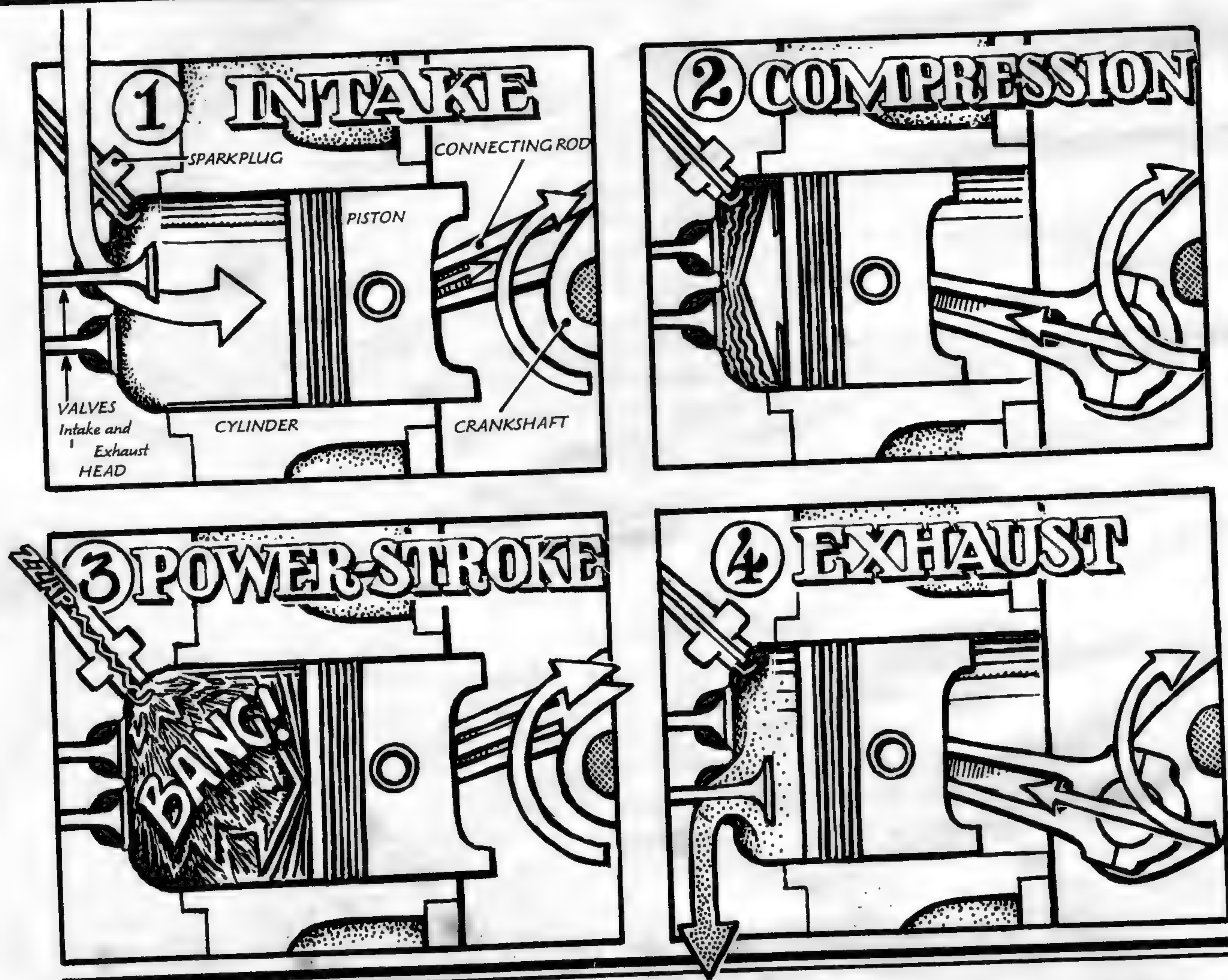
So you see that when the points are closed, the 6 (or 12) volt current goes directly to ground. It is only when the points are open and there's a break in this primary circuit that the coil produces the high voltage (like 30,000 volts) necessary to make the spark jump and be distributed to the spark plug whose turn it is. And this happens over 7,000 times a minute!



Now we have provided the three things necessary to the operation of the engine—fuel-air mixture, spark and compression (the rings and valves)—so let's go through the cycles which make it work.

The **power stroke** takes place when the gas in the combustion chamber (the top of the cylinder and bottom of the head) is burned and forces the piston toward the crankshaft through the connecting rod to the crankshaft throw which translates the sideways motion and power of the moving piston in the cylinder to the round and round motion of the crankshaft. The valves are both closed until the burning is complete at the bottom of the stroke, then the exhaust valve opens and the piston on its way up shoves all of the burned gases out through the exhaust manifold to the muffler and so to the atmosphere. This is the **exhaust stroke**. As the piston comes again to the top of its movement,





the exhaust valve closes, the intake valve opens, and the intake stroke begins. The piston sweeps downward again and sucks the mixture from the intake manifold into the cylinder. As the piston reaches the bottom and the cylinder is full of the gasoline and air mixture, both the intake and exhaust valves close and the piston starts up on the **compression stroke** which jams the mixture up into itself at a ratio of about seven to one as the piston comes to the top again. Just a mite before the piston gets to the top of this stroke, the points open and the plug fires and the mixture burns as, valves closed, the piston is shoved down in the power stroke. There, that's not too hard. Note that I

said **burns** (oxidizes rapidly). When the air-fuel mixture explodes, it causes a pinging noise which indicates that the mixture is being ignited before it is completely compressed and that a change in the timing of the engine is needed. If your engine pings due to a change in altitude or fuel octane, the spark timing should be retarded (slowed down).



The Volkswagen engine is air-cooled, which means that the heat of combustion in the cylinders and heads is carried away by the flow of air. These parts are cast with fins to aid in the removal of heat. A **blower** is provided to force air down through the spaces between the fins to carry the heat to the atmosphere. The entire engine is shrouded by sheet metal to hold

the flowing air to the most efficient path. The blower in Type I and Type II (up to 1971) engines is attached to the back of the generator and is thus driven with the generator by a belt from the crankshaft pulley. The blower is encased in a **fan housing** which is attached to the shrouding to make an air-tight assembly. The cooling air is sucked from the front of the fan housing and blown down



through the cylinders. Also in this fan housing, bolted to the top of the crankcase, is an oil cooler which stands up in the stream of air like a radiator and cools the oil which is pumped through it.

In the Types II (1972 and on), III and IV engines, the fan is attached to the rear of the crankshaft and thus is driven directly. The generator, to the left on Type III's and to the right on Type IV's is driven by a belt. The oil cooler is laid on its side and the resulting flat profile has resulted in the name **pancake engine** for these types.

The generator or alternator is an electrical device which makes six or twelve volt current to charge the battery, and to supply directly the lights when running at night.

The **starter motor** is a simple 6 or 12 volt electric motor used to turn the engine over for starting.

The **voltage regulator** is a box with a lot of wires coming out of it, either on top of the generator or bolted to the side of the engine compartment, which controls the current and voltage to the battery and acts as a relay to operate the lights directly at night.

Structurally the VW Bug uses a bottom pan which is part of the body itself as a frame to bolt the running gear onto the bottom and the seats and things on the top. The Bus is much the same idea but does have a frame to give added strength, as do the other types.

This is very simple but will give you some answers to how the Volkswagen does what it does.

